From Points to Prints: Generating Building Roofprints and Footprints from Airborne LiDAR Data and Initial Outlines

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Abstract

1. Introduction

TODO: Things to mention

- Importance of **building outlines** for various applications (e.g., urban planning, disaster management, etc.)
- Even more applications with **3D building models** (e.g. solar potential, wind modelling, etc.)
- **Several definitions** of building outlines (roofprints vs footprints which are themselves not well defined)
- Algorithms to reconstruct **LoD 2.2 building models** from airborne LiDAR data and initial building outlines (roofer, SimpliCity, etc.)
- Step from Lod 2.2 to LoD 2.3: add roof overhangs to the building models

2. Related Work

TODO: Things to mention

- Related work on roof overhangs
- Related work on roof estimation from airborne LiDAR data?

The topic of roof overhangs reconstruction is not widely covered in the literature, with only a few papers addressing issues related to roof overhangs. The difference between footprints and roofprints is often not acknowledged, with the term outlines often used interchangeably.

The most common method to estimate roof overhangs consists in sweeping vertical planes perpendicularly to the roofprint edges or to the footprint edges depending on what was computed first. Then, a best-fitting plane is determined among these planes with different criteria. In (Panday and Gerke, 2012), a correlation score is computed for each plane, and the best result is kept only if it represents a sharp enough peak compared to its neighbours. For each edge of the footprint, Dahlke et al. (2015) computes the median height on segments parallel to the edge using a precise 2.5D Digital Surface Model (DSM) with a resolution between 5 and 20 cm. Then, they use the inflection point of the height variation as the roofprint edge. In (Frommholz et al., 2017), the roofprint is projected onto the 5 cm resolution DSM and the zero-crossings of the second-order derivative of height variation are used to estimate the size of the roof overhang.

Other methods are proposed by Goebbels and Pohle-Fröhlich (2023) to extend Level of Detail (LoD) 2 models by identifying potential overhang edges from the footprints and computing the size of the overhangs from either oblique images or point clouds. Using obliques images and assuming angles of 45°, they identify the roof overhang in the texture using either edges detection or colour regions, and compute the size of the overhang from this. The method with point clouds extends the roof planes and identify the inliers with a threshold.

Some interesting machine-learning methods were also proposed to compute building outlines from point clouds and could potentially be used to compute both footprints and roofprints. Girard et al. (2020) uses a machine learning model to compute, for each pixel of a RGB aerial image, a classification of building and building edges, as well as a frame field defining tangents and normals of the buildings. Then a multi-step geometric process is used to construct roofprints as polygons. Dai et al. (2025) uses a deep learning model that inputs a point cloud and outputs footprints as a binary raster. The model uses sparse voxel representations for the point cloud and decoder/encoder architectures with a specific 3D attention module. Saadaoui et al. (2025) on the other hand uses an almost full deep learning pipeline to produce roofprints as polygons, getting rid of the constraints of rasterization. A first model identifies building pixels, followed by a residual autoencoder to regularize the segmentation, and finally a lightweight CNN that extracts building corners that can be used for polygonization.

Regarding data, to our knowledge, there is a lack of datasets featuring Airborne Laser Scanning (ALS) data combined with both footprints and roofprints, making it difficult to evaluate the results of the methods that we propose. The only mention of a similar dataset that we found is Dai et al. (2025) stating that they will release a dataset with more than 3000 building footprints based on the ALS dataset called DALES (Varney et al., 2020). However, it is not yet available at the time of writing this.

3. Methodology

TODO: Things to mention

- First roofprint then footprint because roofprints are easier to estimate from airborne LiDAR data (less occlusions, more points, etc.)

The different parts of the proposed method are shown in Figure 1. The method aims at generating two 2D polygons for each building — one for the roofprint and one for the footprint — from two main inputs: ALS data and an initial building outline. Due to the nature of ALS data, the density of points on roof surfaces is significantly higher than on façades, which is the reason why we first focus on the estimation of roofprints and then footprints.

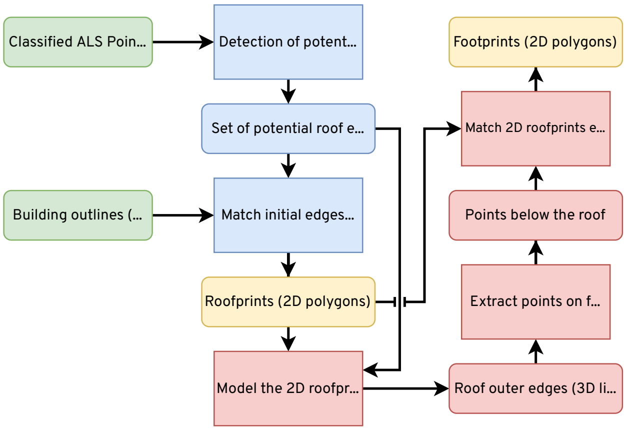

Figure 1. Overview of the proposed pipeline for generating building roofprints and footprints.

The pipeline can be summarized as:

1. Identify points corresponding to roof edges,
2. Use these points to make a roofprint,
3. Identify points corresponding to façades and ground below the roof,
4. Use the points to make a footprint.

3.1 Input data

TODO: Things to mention

- Airborne LiDAR data (point cloud):
  - High point density (e.g., 10-20 points/m^2) and high precision and accuracy (e.g., 10-20 cm).
  - Semantic segmentation between vegetation and non-vegetation points
- Initial building outlines (e.g., from cadastral data, building detection algorithms, etc.) with:
  - Correct topology and shapes (angles between walls)
  - Approximately correct size
  - Poor georeferencing (displacement of up to a few meters)
- Point cloud trajectory (can be automatically computed using multi-echo rays)

This method expects a high-density ALS point cloud as an input (at least 10 points/m^2). This point cloud should be semantically segmented and contain at least a separate class for vegetation and ground. It should also contain enough information to isolate the flight strips and order the points in acquisition order. A GPS time attached to each point is sufficient for this purpose, and an ID specific to each flight strip can simplify the process. Finally, the trajectory of the sensor is also necessary, but it can be computed automatically using for example the multi-echo pulses of the point cloud. [HOW TO CITE THE ALGO OF WU TENG?]

The second input consists in building outlines. These building outlines can be either footprints, roofprints or a mix of both. They are expected to be roughly the correct size and roughly in the right spot up to a few metres. But their main characteristic is that their overall shape and topology, as well as the angles of their edges, are expected to be perfect. *Maybe an illustration of the expected input with an example from BD TOPO would help?* These two requirements come from the constraints imposed on the optimization process: never flip an edge and never rotate an edge. Section 3.3 gives more details about the optimization process.

3.2 Identification of roof edge points

TODO: Things to mention

- Point cloud topology (flight strips, scan lines, pulses)
- Computing maximum vertical gap with neighbours in same and adjacent scan lines
- Displacement of single-echo points to be closer to the roof edge

3.3 Construction of roofprints

TODO: Things to mention

- Optimization of all connected edges together to preserve topology
- Description of the edge shifting algorithm
- Energy function to minimize which combines two terms:
  - Proximity to the roof edge points (requires to define the proximity score and the inward direction)
  - Similarity to the initial edges (to preserve the shape and size of the building)
- Multiple iterations to converge to the final roofprint

3.4 Identification of wall and ground points

TODO: Things to mention

- Transfer of the 2D roofprints into 3D by identifying the best 3D segments corresponding to each 2D edge of the roofprint
- Extract points below and towards the inside (points on walls or on the ground indicating the presence of a roof overhang)

3.5 Construction of footprints

TODO: Things to mention

- Optimization of each edge individually
- Corresponds to the estimation of the roof overhang for each edge of the roofprint
- Other energy function to minimize which combines two terms to try to use both the wall points and the ground points (in 2D):
  - Proximity to the wall and ground points
  - Absence of points more towards the inside

4. Results

TODO: Things to mention

- For now mostly qualitative results are possible. I need to find different areas with different types of buildings to test it.
- Quantitative evaluation one a limited number of buildings for which I create manually the roofprints and footprints would be great

5. Discussion

TODO: Things to mention

- Strengths of the method:
  - Preserves the quality of the initial building outlines
  - Can take any kind of outline as an input
  - Robust to outliers and to partial occlusions thanks to the preservation of the initial building outlines
- Limitations of the method:
  - Relies a lot on the quality of the initial outlines
  - Identification of points of interest requires to exclude vegetation points because they would otherwise be picked
  - Struggles with adjacent buildings for the 3D part
  - A significant number of parameters to set (weights for the energies, distances for the proximity scores, threshold for the identification of roof edge points and for inward directions, etc.)
- Potential improvements:
  - Training a semantic segmentation machine learning model to identify wall points, roof points and ground points
  - Automating the parameter tuning process
  - Give more freedom to the edge shifting algorithm to improve the outlines when possible (incorrect topology, slightly wrong angles, etc.)

An interesting property of most buildings that is not used by this method is the symmetry. Panday and Gerke (2012) used it to assume that roof overhangs are the same on opposite sides when the roofs are slanted, which could help computing better estimations. If there are points on both sides, it increases the amount of data that can be used and an average of both sides can be a better estimation, while if there are points on only one side, it can be used to make a reasonable guess of the overhang on the opposite side.

6. Conclusions

TODO: Things to mention

References

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